

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MEC 1C 01—MATHEMATICAL ECONOMICS

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 15

Maximum : 15 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 1C 01—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following are NOT true ?

- (A) $f(x) = ax^n$ implies $f'(x) = anx^{n-1}$.
- (B) $f(x) = 4x^5 - 3x^2$ implies $f'(x) = 20x^{-5} - 6x^{-2}$.
- (C) $f(x) = 4x + 3/x^2$ implies $f'(x) = 4 - 6/x^3$.
- (D) $f(x) = \ln(x)$ implies $f'(x) = x^{-1}$.

2. Which of the following statements are (in general) true ?

- (A) Marginal cost (MC) is minimised where MC = Average Variable Cost (AVC).
- (B) Total Cost (ATC) is minimised where MC = ATC.
- (C) Average Variable Cost (AVC) is minimised where MC = AVC.
- (D) Total revenue is maximised where MC = Marginal Revenue (MR).

3. For the function $Q = AK^aL^b$ which of the following statements are NOT true ?

- (A) $dQ/dL = AbK^aL^{b-1}$.
- (B) Marginal Product of Labour (MPL) = $AaK^{a-1}L^b$.
- (C) Marginal Product of Capital (MPK) = aQ / K .
- (D) Marginal rate of substitution of capital for labour (MRS) = $|dK / dL|$.

4. The law which studies the direct relationship between price and quantity supplied of a commodity is :

- (A) Law of demand.
- (B) Law of variable proportion.
- (C) Law of supply.
- (D) None of the above.

5. In case of perfectly inelastic supply the supply curve will be :

- (A) Rising.
- (B) Vertical.
- (C) Horizontal.
- (D) Falling.

6. At what point does total utility starts diminishing ?
- (A) When marginal utility is positive.
 - (B) When it remains constant.
 - (C) When marginal utility is increasing.
 - (D) When marginal utility is negative.
7. Few sellers is the feature of :
- (A) Monopoly.
 - (B) Oligopoly.
 - (C) Perfect competition.
 - (D) Monopolistic competition.
8. Supply curve of a perfectly competitive firm is :
- (A) Vertical.
 - (B) Upward sloping.
 - (C) Horizontal.
 - (D) Downward sloping.
9. Suppose the supply for product A is perfectly elastic. If the demand for this product increases :
- (A) The equilibrium price and quantity will increase.
 - (B) The equilibrium price and quantity will decrease.
 - (C) The equilibrium quantity will increase but the price will not change.
 - (D) The equilibrium price will increase but the quantity will not change.
10. If the demand for agricultural products is inelastic :
- (A) As the prices decrease, the revenues earned by producers increase.
 - (B) As the prices decrease, the revenues earned by producers decrease.
 - (C) Rising prices do not lead to differentiation in producers' incomes.
 - (D) The percentage decrease in prices is lower than the percentage increase in demand.
11. If the demand curve for product A moves to the right, and the price of product B decreases, it can be concluded that :
- (A) A and B are substitute goods.
 - (B) A and B are complementary goods.
 - (C) A is an inferior good, and B is a superior good.
 - (D) Both goods A and B are inferior.

12. If a price increase of 50% results in an increase in the quantity supplied of an economic good from 10 to 20 pieces, calculate the co-efficient of price elasticity of supply :
- (A) $1/4$. (B) $1/2$.
(C) 1. (D) 2.
13. An economic agent contracts a loan of 15.000 lei, which he will repay in three equal annual installments. What will be the total interest paid, knowing that the annual interest rate is 12% per year ?
- (A) 3.600 lei. (B) 1.800 lei.
(C) 5.400 lei. (D) 1.500 lei.
14. Which of the following statements is false :
- (A) Perfect competition involves many sellers of standardized products.
(B) Monopolistic competition involves many sellers of homogeneous products.
(C) The oligopoly involves several producers of standardized or differentiated products.
(D) Monopoly involves a single product for which there are no close substitutes.
15. If the price of coffee falls by 8% and the demand for Tea declines by 2%. The cross price elasticity of demand for Tea is :
- (A) 0.45. (B) 0.25.
(C) +0.44. (D) - 0.30.

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

MEC 1C 01—MATHEMATICAL ECONOMICS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. Define Law of Demand.
2. What is meant by Cross elasticity of demand ?
3. Define MRS_{xy} .
4. Define AFC.
5. Explain the meaning of Short run costs.
6. What is meant by a Point of inflexion ?
7. What is an Indifference map ?
8. Explain the term Shift in demand curve.
9. Explain the meaning of Budget line.
10. What is meant by Constrained optimization?
11. If $TC = 5Q^2 + 12Q + 14$, find MC.
12. Define the term Consumer equilibrium.

(8 × 3 = 24 marks)

Turn over

Section B

*Answer at least **five** questions.*

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Derive the relation between MR, AR and elasticity of demand.
14. What is ordinal utility of demand ?
15. State the law of equi-marginal utility. If the utility function is $U = f(q_1, q_2)$ and the budget equation is $M = p_1q_1 + p_2q_2$, derive the law of equi-marginal utility.
16. Explain the properties of indifference curves.
17. Explain the conditions for the optimization of the multivariable functions.
18. Assume a four sector economy, where $Y = C + I + G + (X - M)$, $C = C_0 + bY$, $I = I_0 + aY$, $G = G_0$, $Z = Z_0$. Find the equilibrium level of income in terms of general parameters.
19. What is marginal productivity ? Given the production function $Q = AL^aK^b$, show that marginal productivity of labor and capital depends on capital (K) – labor (L) ratio.

(5 × 5 = 25 marks)

Section C

*Answer any **one** question.*

The question carries 11 marks.

20. Explain the significance of Lagrange multiplier and optimize the function $3x^2 - 2xy + 6y^2$ subject to the constraint $x + y = 36$ using the Lagrange multiplier.
21. Explain cardinal utility analysis of demand. Derive consumer equilibrium using cardinal utility method.

(1 × 11 = 11 marks)